

COOCK | OPEN STAD



VALUABLE ENVIRONMENTAL INSIGHTS
THROUGH AN OPEN CITY APPROACH

Mobile Air Quality in Antwerp on postal vans
Jelle Hofman (IMEC-NL)

10.00 – 10.15

Mobile monitoring: A general introduction, Martine Van Poppel (VITO)

10.15 – 10.35

Mobile monitoring using airQmap: data analysis and lessons learned, Jan Peters/Martine Van Poppel (VITO)

10.35 – 10.55

Mobile Air Quality in Antwerp on postal vans, Jelle Hofman (IMEC-NL)

10.55 – 11.15

Road traffic noise as proxy for traffic: multidisciplinary applications, policy support and smart city potential, Luc Dekoninck (UGent)

11.15 – 11.35

Denoising of opportunistic mobile measurements, Dick Botteldooren (UGent)

11.35 – 11.55

Mobile Measuring of Electromagnetic Fields, Wout Joseph (UGent)

11.55 – 12.00

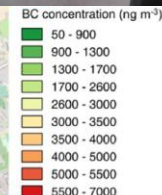
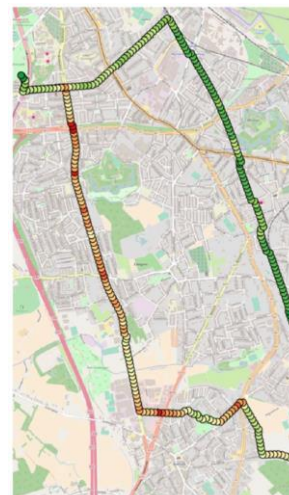
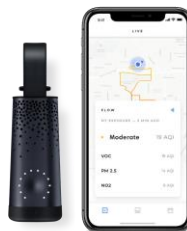
Wrap-up

MOBILE AIR QUALITY USE CASES

MOBILE SENSING

■ Why?

- Personalizing environmental data: **personal exposure**
 - More representative
 - More personal → impact behaviour
 - Awareness raising
- Higher spatial resolution: **air quality mapping**
 - Covering wider area (“at every doorstep”)
 - Mapping, Hotspot detection, policy support,...
 - At cost of temporal resolution
- Logistics
 - Fleet maintenance/calibration
- Complementary information (GPS)
 - Desired cycling routes
 - Traffic dynamics
 - ...



0 1 km

<https://www.sciencedirect.com/science/article/abs/pii/S0013935118301245?via%3Dihub>



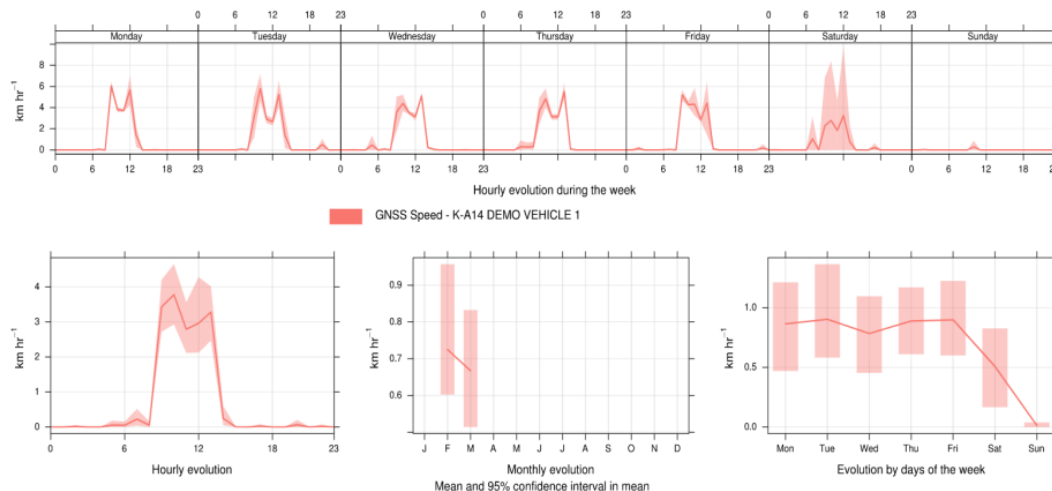
CHALLENGES

■ Data Quality

- Movement (turbulence/vibration)
- Comparability (positioning, sampling conditions, standardisation, ...)
- Evaluation protocol

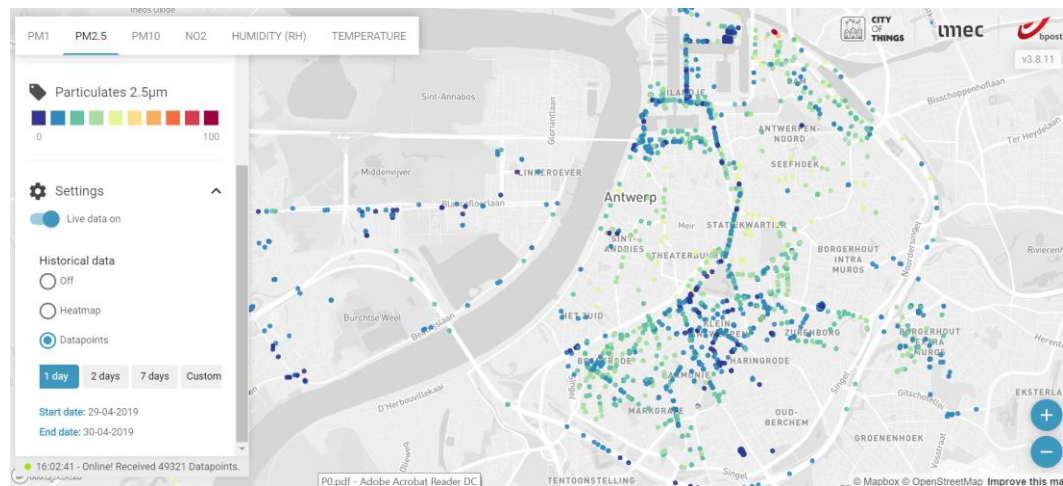
■ Representativity

- Monitoring resolution (~platform)
 - 20 km/h $\rightarrow$ 55 m (10 sec)
 - 50 km/h $\rightarrow$ 139 m (10 sec)
 - LoRa vs LTE-M/NB-IoT
- Spatial data coverage (mapping)
 - Limited temporal resolution!
 - Coverage factor (%)
 - Temporal correction
- Standardisation



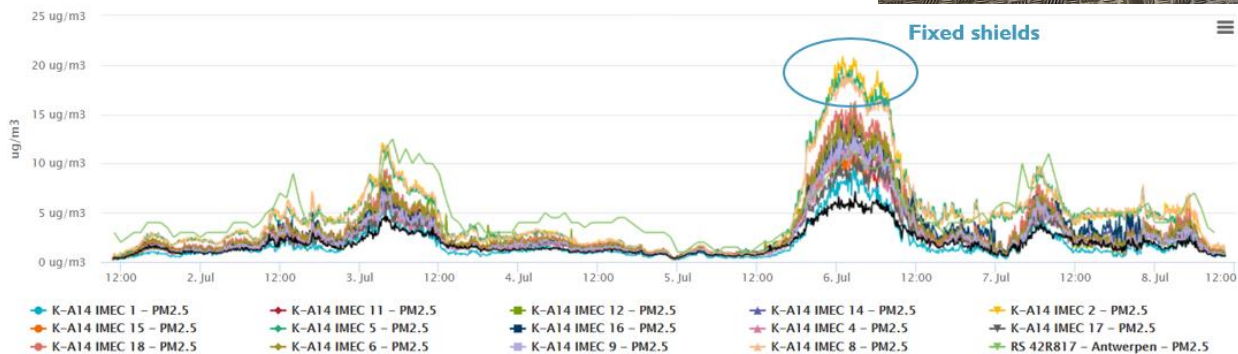
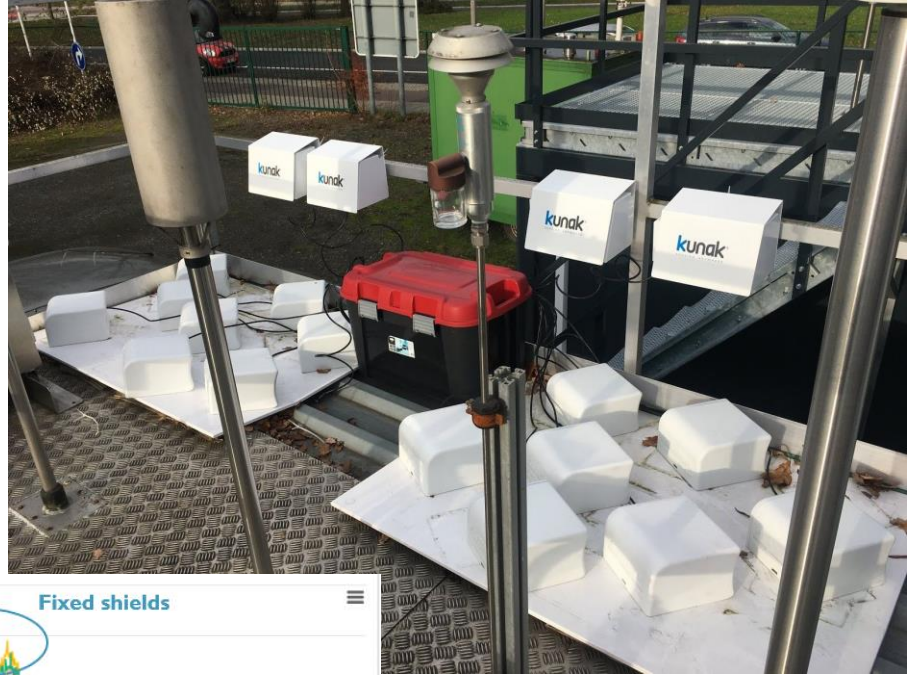
POSTAL VANS

- Fine-grained air quality mapping
- 20 sensors on bpost vans, Antwerp (BE)
 - LTE-M
 - Dedicated housing
 - Power: car battery (12V ~100mA)
- v1 (LoRa) – v2 (Kunak, LTE-M)



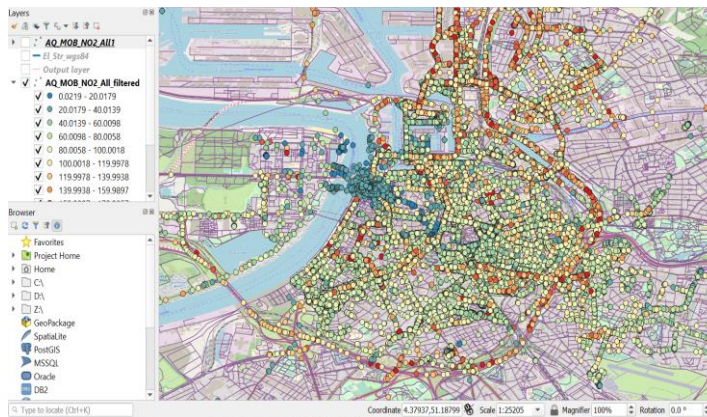
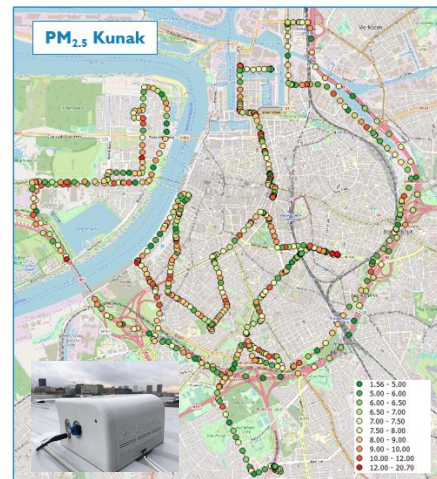
POSTAL VANS

- **Impact housing: PM vs NO₂**
- Mobile performance
 - No protocol/REF monitors
- Recurrent calibration



POSTAL VANS

- Impact housing
- **Mobile performance**
 - No protocol/REF monitors
 - Drive-by comparison against REF
 - Comparison mobile instruments (FIDAS Frog)
 - Data exploration
- Recurrent calibration



R803



R805



R802

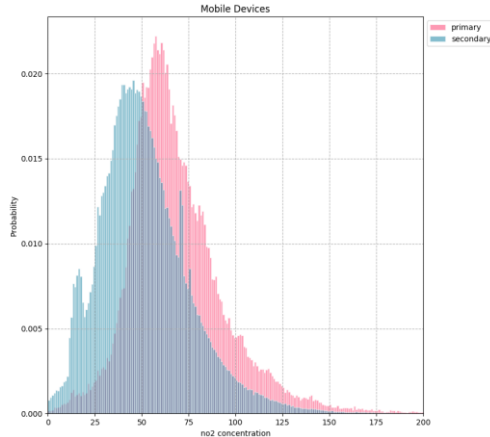


R804

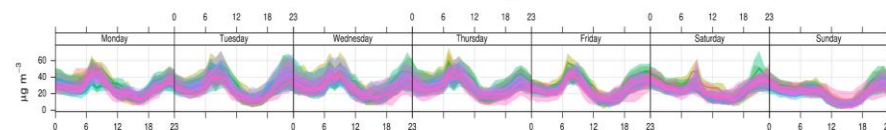


POSTAL VANS

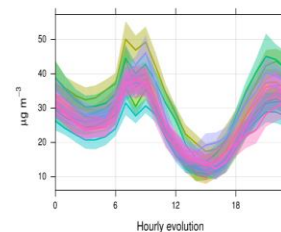
- Impact housing
- **Mobile performance**
 - No protocol/REF monitors
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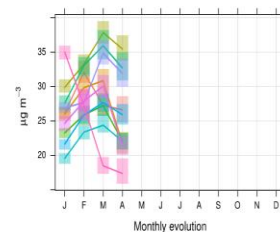
Time variation several devices: NO₂ GCc AVG1H



Hourly evolution during the week

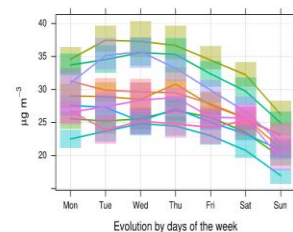


Hourly evolution



Monthly evolution

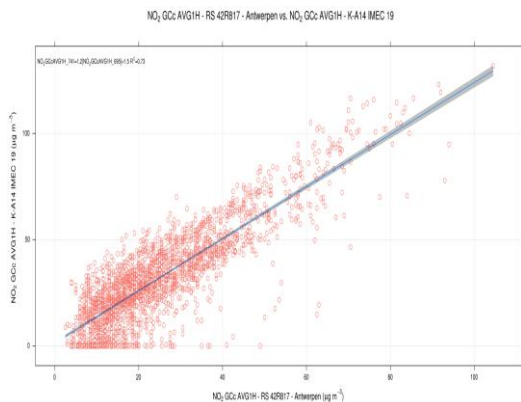
Mean and 95% confidence interval in mean



Evolution by days of the week

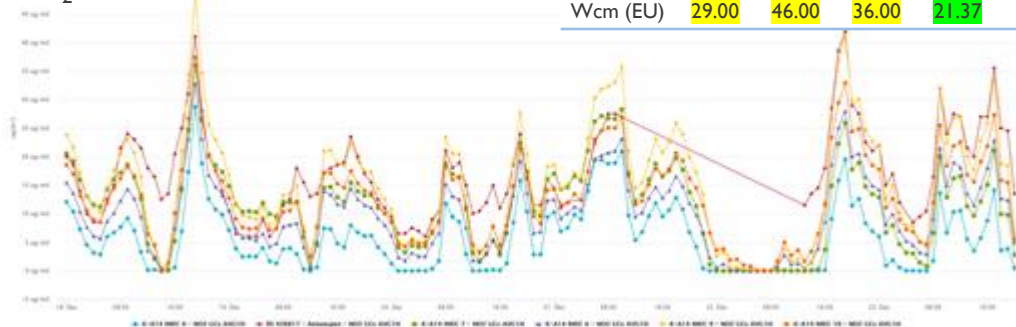
POSTAL VANS

- Impact housing
- Mobile performance
 - No protocol/REF monitors
- **Recurrent calibration**

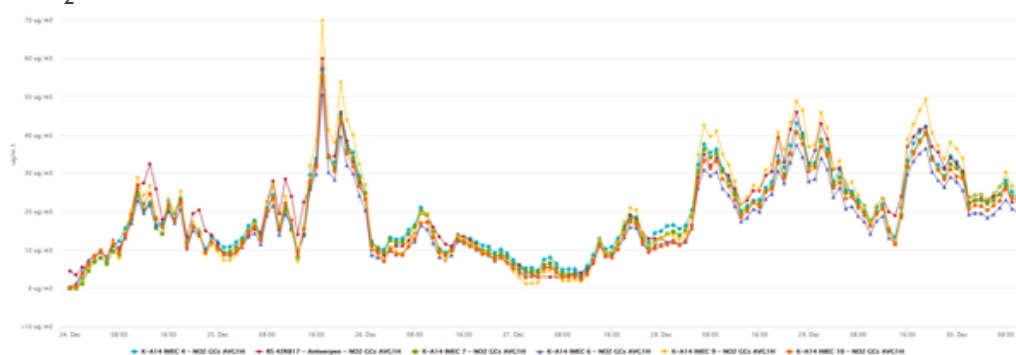


	IMEC 4	IMEC 6	IMEC 7	IMEC 9	IMEC 10
R ²	0.82	0.90	0.81	0.90	0.93
MAE	4.50	6.75	5.28	3.50	3.77
RMSE	6.64	8.11	7.45	5.79	5.48
Wcm (EU)	29.00	46.00	36.00	21.37	23.00

NO₂ BEFORE



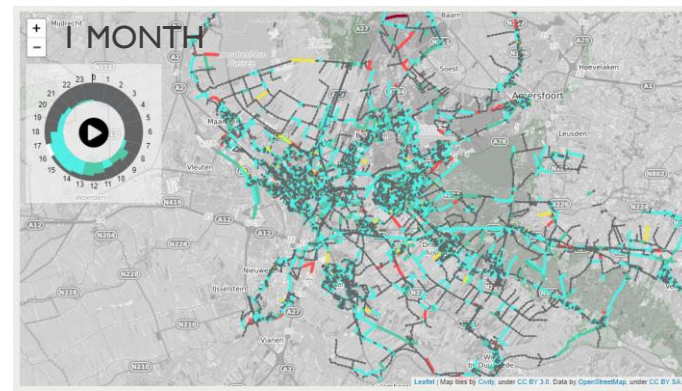
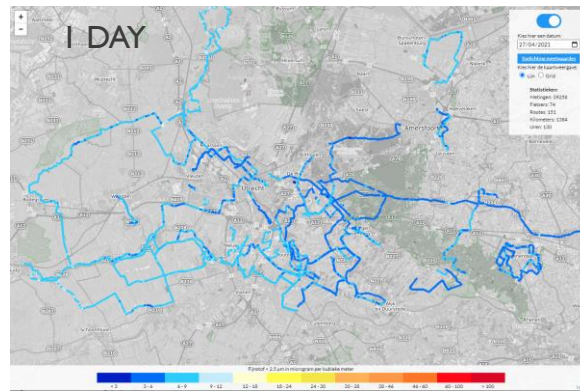
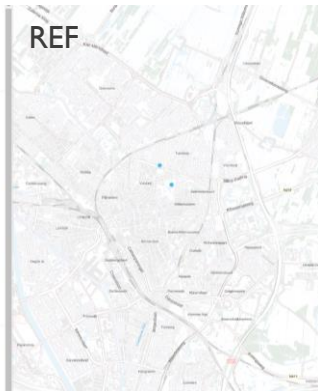
NO₂ AFTER



R²= 0.88-0.97, MAE = 2.18 - 3.38, RMSE = 2.96 – 4.24

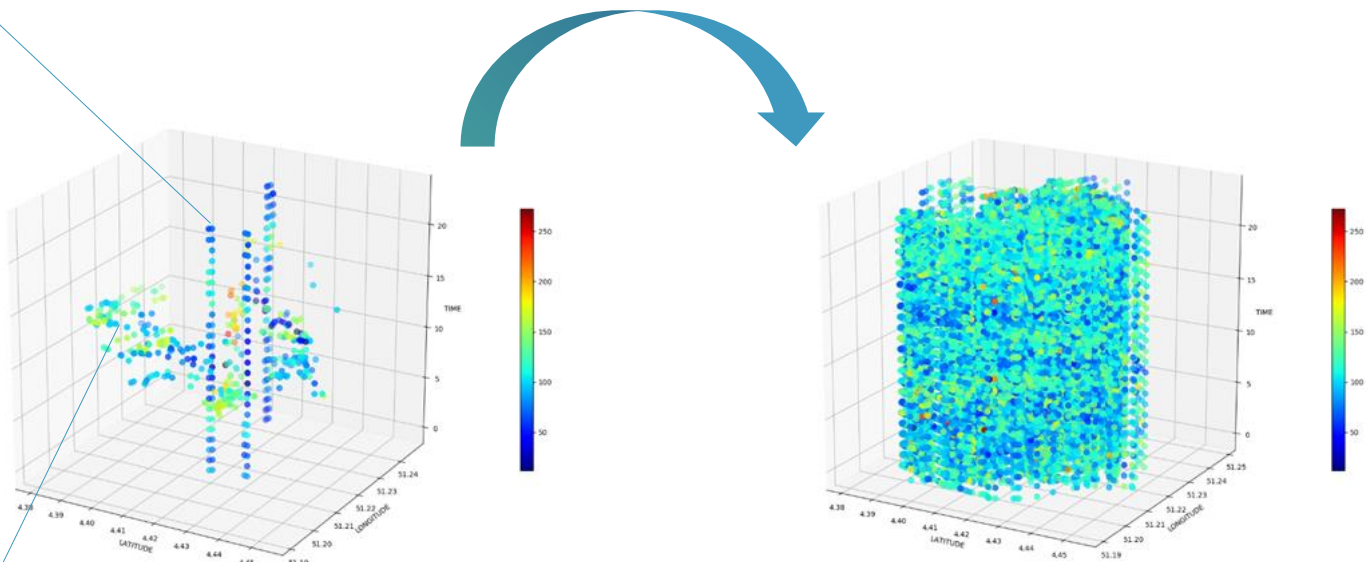
SNUFFELFIETS

- Snuffelfiets, Utrecht (NL)
 - Sensirion SPS30 sensor mounted on bicycles ($PM_{2.5}/PM_{10}$)
 - Battery/LTE-M
 - Calibrated by RIVM based on mass correction factor (REF/sensor) derived at air quality stations
- From personal trajectory → air quality map



<https://snuffelfiets.nl/>

AIR QUALITY INFERENCE



https://vloca-kennishub.vlaanderen.be/vloca-kennishub/Cooock_Open_Stad/IoT_and_Air_Quality_in_Cities_part_2

AIR QUALITY INFERENCE

- Validation inference models
 - FAIRMODE temporal validation

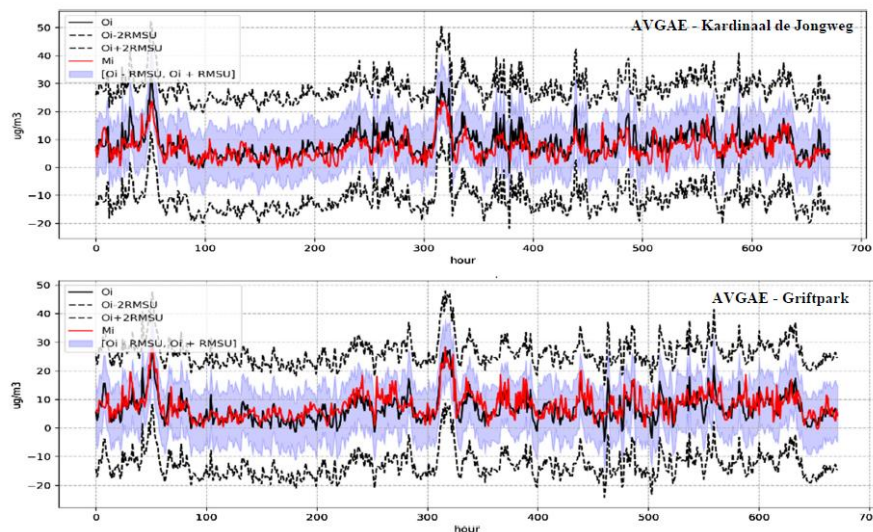


Table 1. Temporal validation statistics for the AVGAE and GRF model at two different regulatory stations in Utrecht (NL10636 Kardinaal de Jongweg, NL10643 Griftpark).

Model	Station	MAE	MBE	RMSE	IA	Acc	Corr	NMB	NMSD	MQI
AVGAE	NL10636	3.14	1.79	3.99	0.82	0.64	0.72	-0.21	-0.07	0.20
AVGAE	NL10643	2.83	0.15	3.73	0.75	0.58	0.63	-0.02	-0.35	0.19
GRF	NL10636	2.88	1.99	3.8	0.79	0.67	0.77	-0.23	-0.34	0.43
GRF	NL10643	2.74	-1.05	3.49	0.79	0.6	0.73	0.15	-0.37	0.44
AVGAE	Avg	2.99	0.97	3.86	0.79	0.61	0.68	-0.12	-0.21	0.20
GRF	Avg	2.81	0.47	3.65	0.79	0.64	0.75	-0.04	-0.36	0.44

https://vloca-kennishub.vlaanderen.be/vloca-kennishub/Coock_Open_Stad/IoT_and_Air_Quality_in_Cities_part_2

OUTLOOK

- (Opportunistic) mobile monitoring
 - Personalizing environmental data
 - More representative insights depending on applied carrier (cyclists, toddlers, vehicles,...)
 - Increase spatial coverage
- Standardisation – Validation!
- Validation of mobile sensor solution
 - Fixed setting: sensor performance
 - Validation solution: e.g. FAIRMODE validation: sensors + model